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Experiment No: 4 **Date: / /**

AIM : Controlling LED using push button with ESP32 (Digital Read)

OBJECTIVE: The main objective of this experiment is,

A. To read the Push button State Using Serial Monitor

B. To turn ON an LED when a button is pressed, OFF when button is released.

APPRATUS: Arduino UNO / ESP32 , LED , PUSH BUTTON , Jumper wires , Breadboard

DESCRIPTION:

In this experiment we simply hook up 5 volts to one side of a button and to the other side of the button we connect pin 12. When you press the button it completes an electrical connection, pin 12 will "see" the 5 volts and if we digitalRead() at pin 12, it will report HIGH. During the times the button is not being pressed, pin 12 reports LOW in PULLDOWN .

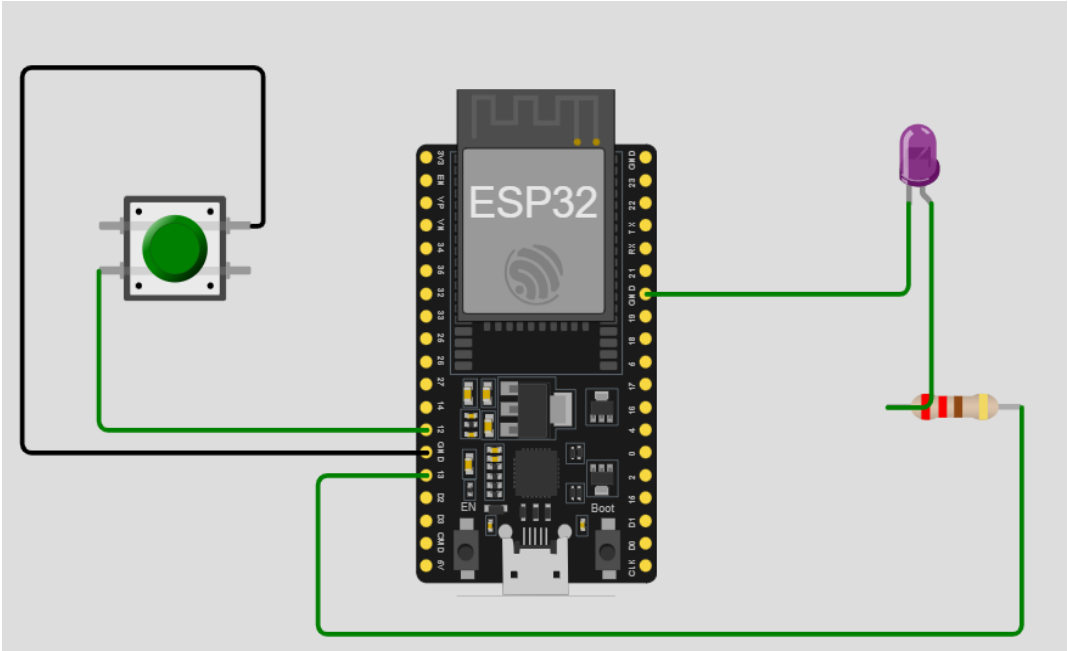
We also have pin 12 connected to ground at all times through a resistor. This is because when we read values at pin 12, we want to get either a HIGH or a LOW reported. If pin 12 is not connected to ground, then when the button is not being pressed it becomes what is called a floating pin – it's not connected to anything. Floating pins on the Arduino are fine for the most part – unless you are trying to record an input from them – then they are bad, and can give you spurious information.

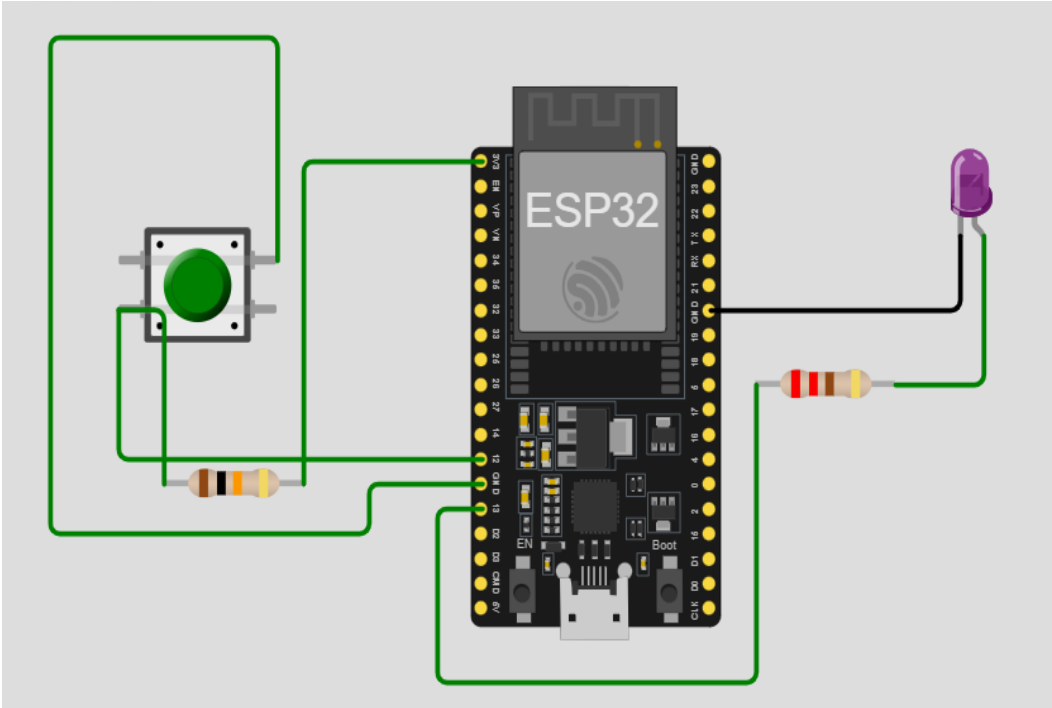
In second case ,if we connect GND to one side of a button and other side of the button is connected with pin 12. We press the button it completes an electrical connection, pin 12 will "see" the GND and if we digitalRead() at pin 12, it will report LOW. During the times the button is not being pressed, pin 12 reports HIGH in PULLUP.

With the internal pull-up enabled, the input logic is inverted: the pin reads HIGH when the button is open and LOW when the button is pressed. This is a simple and common way to interface buttons without extra components.

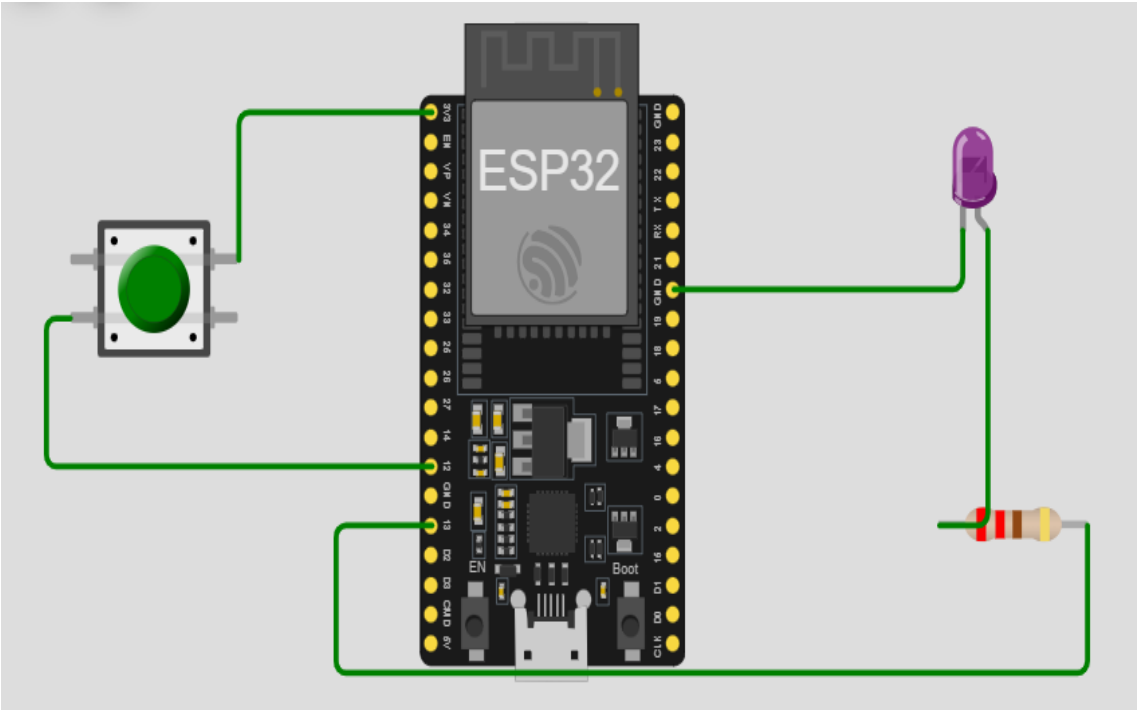
INTERFACING DIAGRAM:

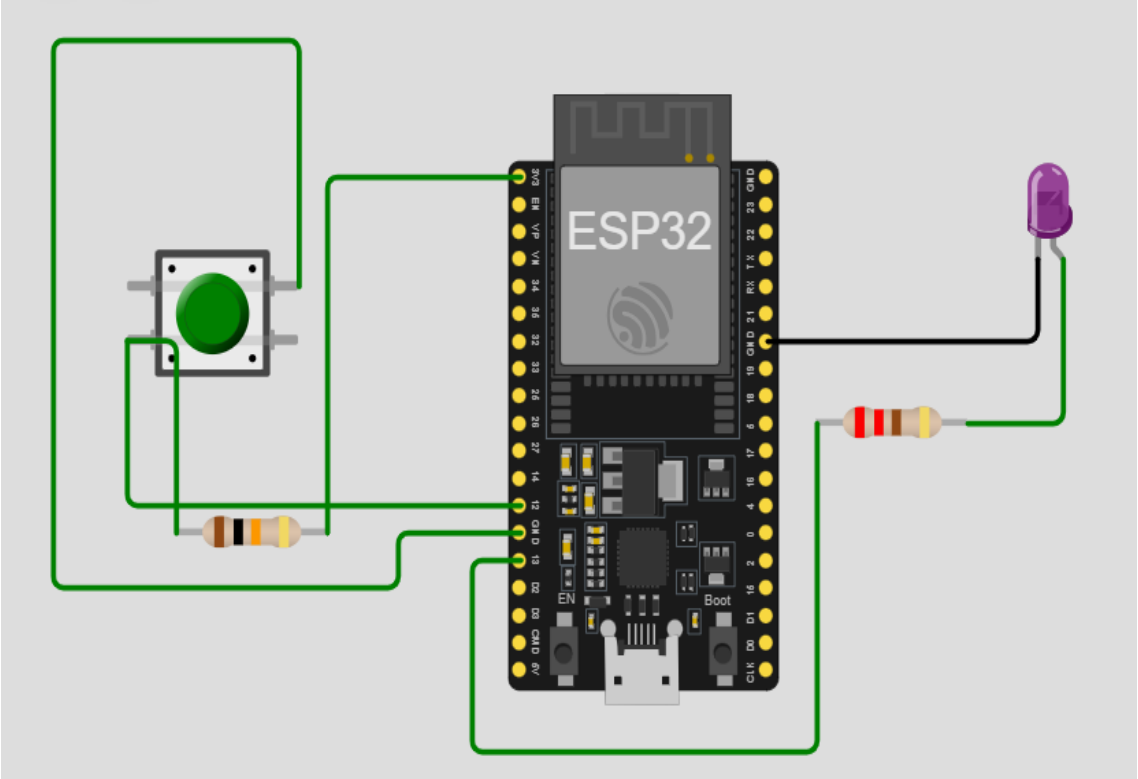
Pull UP





Pull Down





PROCEDURE:

- Connect one of the ESP32 GND pins to one of the long power rails on the breadboard –this will be the ground rail.
- Connect the short leg of the LED to this same ground rail on the breadboard, then connect the long leg to a row on the breadboard.
- Connect the 220-ohm resistor from pin 13 to the same row that you have the longleg of the LED attached.
- Place the pushbutton on the breadboard. Most buttons will straddle the centre trench on the breadboard.
- Connect a jumper wire from the 5-volt/GND pin to one side of the pushbutton, depending on pullup and pull down.
- Connect a jumper wire from pin 12 to the other side of the pushbutton.
- By connecting the **10kΩ resistor** between **GND** and the button terminal (the same side connected to the GPIO pin), you have created a standard **Pull-down** circuit.
- For pullup, The Button: Connect one terminal of the push button to the 3.3V pin. Connect the other terminal to GPIO 12.The 10kΩ Resistor: Connect one end to GPIO 12 (the same junction as the button) and the other end to the GND rail.

PROGRAM:

RESULTS:

CONCLUSION:

SAMPLE VIVA-VOCE QUESTIONS:

1) How does push button work in ESP32?
2) What are the types of push button?
3) Does push button need resistor?
4) What are the applications of push button?
5) How many pins are there for push button?